

Assignment: GenAI-Enhanced Advanced Data Structures

Weight: 10% of total module mark

Type: Individual assignment

Submission: Code + Short Technical Report

Estimated effort: ~20 hours

1. Assignment Context & Rationale

Modern software engineers increasingly work **with** AI-assisted development tools rather than in isolation. This assignment is designed to help students:

- Deepen their understanding of **advanced data structures in Java**
- Critically and responsibly integrate **Generative AI (GenAI)** tools into software development
- Reflect on the **benefits, limitations, and risks** of AI-assisted coding

The assignment is inspired by real-world challenges faced by MSc Computer Science students when implementing complex data structures—such as performance optimisation, correctness, debugging, and design trade-offs.

Generative AI **must form part of your learning and development process**, but **you remain fully responsible for the final design, correctness, and explanation of your solution**.

2. Assignment Overview

You will:

1. **Select ANY ONE advanced data structure** (see list below)
2. **Implement it in Java from scratch** (do NOT use Java's built-in implementations)
3. **Use Generative AI tools** to assist with:
 - Design exploration
 - Code generation or refinement
 - Debugging or optimisation
4. **Critically reflect** on how GenAI influenced your design decisions and learning

3. Choice of Advanced Data Structure (Pick ONE)

You may choose **any ONE** of the following (or propose an equivalent, subject to approval):

- Self-balancing trees
 - AVL Tree
 - Red-Black Tree
- Advanced hash-based structures
 - Custom HashTable with collision strategies
 - Cuckoo Hashing
 - Hopscotch Hashing
- Priority & indexing structures
 - Fibonacci Heap
 - Binomial Heap
- Graph-based structures
 - Adjacency List / Matrix with optimised operations
 - Union-Find with path compression & ranking
- Range & spatial structures
 - Segment Tree
 - Interval Tree
 - KD-Tree

4. Required Use of Generative AI

You **must use at least THREE Generative AI tool**, such as:

- ChatGPT / GPT-based tools
- GitHub Copilot
- Tabnine
- SpellBox
- Other approved GenAI coding assistants

Acceptable Uses

- ✓ Generating initial design ideas
- ✓ Producing draft code snippets
- ✓ Explaining algorithms
- ✓ Identifying bugs or performance issues
- ✓ Suggesting optimisations

Not Acceptable

- ✗ Submitting AI-generated code without understanding it
- ✗ Copy-pasting without modification or explanation
- ✗ Claiming AI-generated work as entirely your own

You must **document and reflect on how GenAI was used**.

5. Implementation Requirements

Your Java implementation must include:

- Core operations relevant to your data structure
(e.g. insert, delete, search, update, rebalance, etc.)
- Clear object-oriented design
- Meaningful comments and documentation
- A **small test harness** demonstrating correctness and behaviour
- Basic performance discussion (e.g. time complexity expectations)

6. Reflective & Technical Report (≈ 1,000–1,200 words)

Submit a short report covering:

A. Data Structure Design (≈ 300 words)

- Why you chose this data structure
- Key design decisions
- Trade-offs involved

B. Use of Generative AI (≈ 400 words)

- Which GenAI tools you used
- How they supported your work (design, coding, debugging, optimisation)
- What worked well and what did not
- Where human judgement was essential

C. Critical Reflection (≈ 300 words)

- Did GenAI improve your understanding?
- Did it introduce risks or misconceptions?
- How would you use (or limit) GenAI in future software engineering work?

7. Submission Checklist

You must submit:

1. **Java source code** (.java files or GitHub link)
2. **Short report** (PDF)
3. **AI usage declaration** (included in report)

Marking Scheme (10%)

Criterion	Weight
Correctness & completeness of data structure	30%
Quality of Java design & implementation	20%
Effective & responsible use of GenAI	20%
Critical reflection on AI use	20%
Clarity & professionalism of report	10%

Academic Integrity & AI Use Policy

- Use of GenAI is **explicitly allowed and encouraged**
- You must **acknowledge AI assistance**
- You are responsible for **all submitted content**
- Plagiarism or undisclosed AI use will be penalised

Learning Outcomes Addressed

By completing this assignment, you will be able to:

- Implement advanced data structures in Java
- Analyse algorithmic trade-offs and performance
- Integrate GenAI tools into software development responsibly
- Critically evaluate AI-assisted programming practices